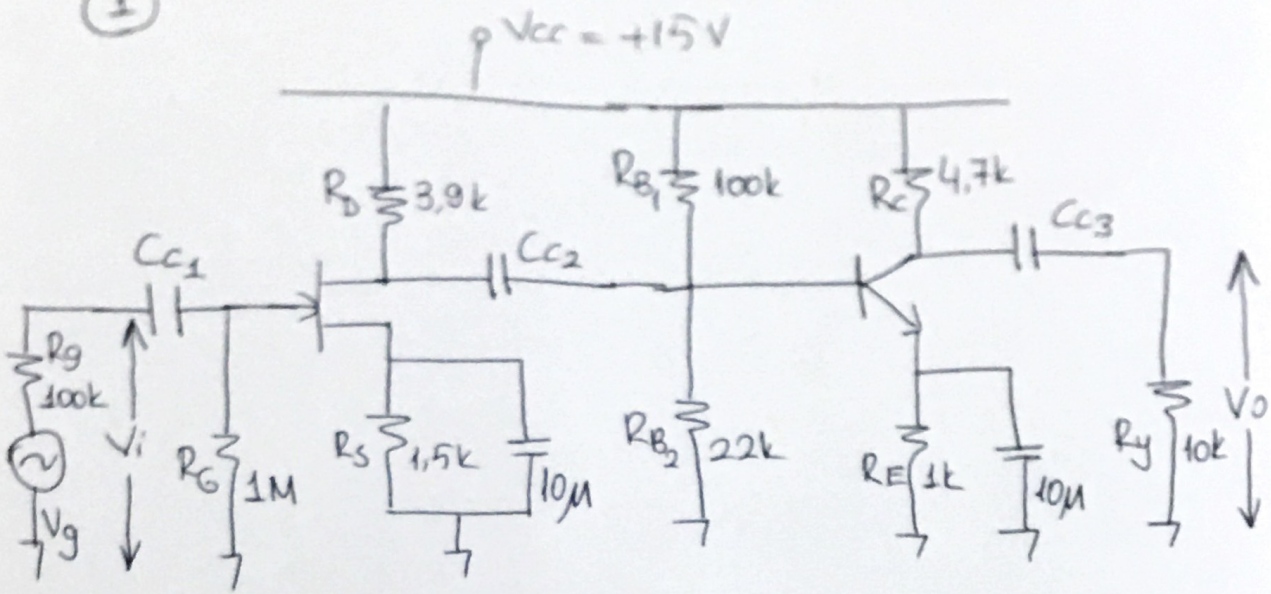


1



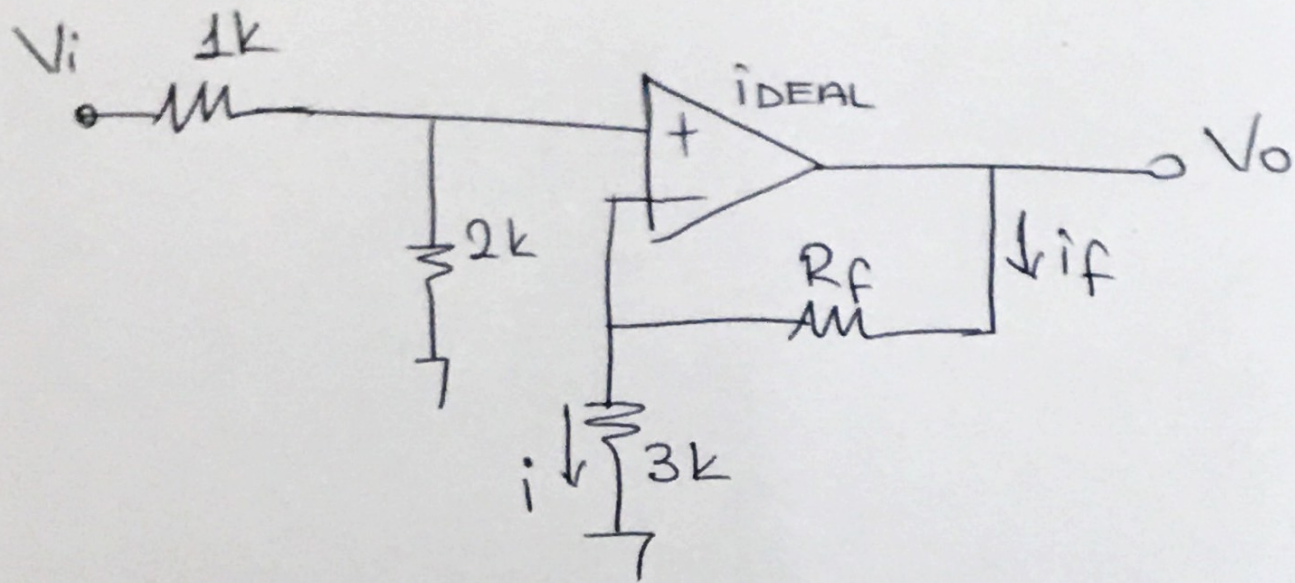
$I_{DSS} = 6 \text{ mA}$, $V_p = -3 \text{ V}$, $h_{fe} = h_{FE} = 200$, $V_{BE} = 0.7 \text{ V}$
 $1/h_{oe} = 50 \text{ k}$ $C_{gs} = 3 \text{ pF}$, $C_{ds} = 1 \text{ pF}$, $C_{gd} = 3 \text{ pF}$
 $C_{be} = 80 \text{ pF}$, $C_{bc} = 3 \text{ pF}$, $C_{ce} = 6 \text{ pF}$

a) $K_v = \frac{V_o}{V_i} = ?$

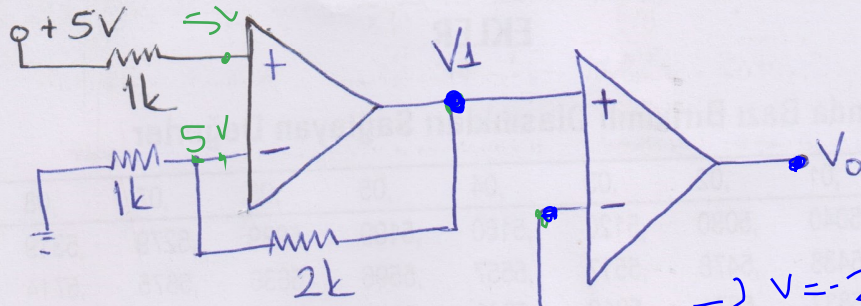
b) Alt kesim frekansının 10 Hz olabilmesi için,
 $C_{c1}, C_{c2}, C_{c3} = ?$

c) Üst kesim frekansı nedir?

② Aşağıda verilen ideal OPAMP'ın kazancının 5 olabilmesi için R_f ne olmalıdır?



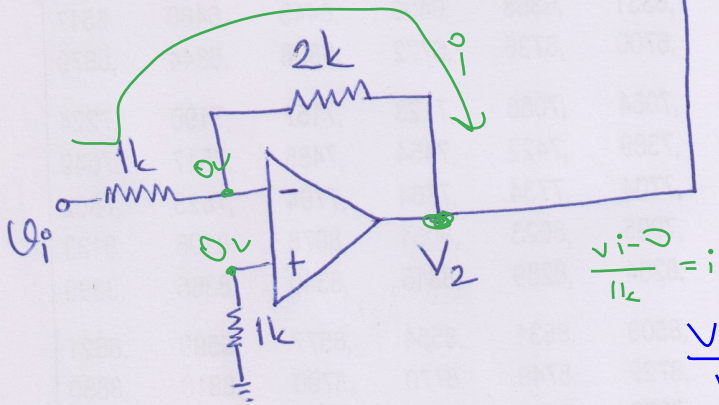
3



$$-2V = -2V$$

$$V_2 = V_1$$

$$\frac{-V_2}{2k} = \frac{1V}{1k} \quad \boxed{V_2 = -2V}$$



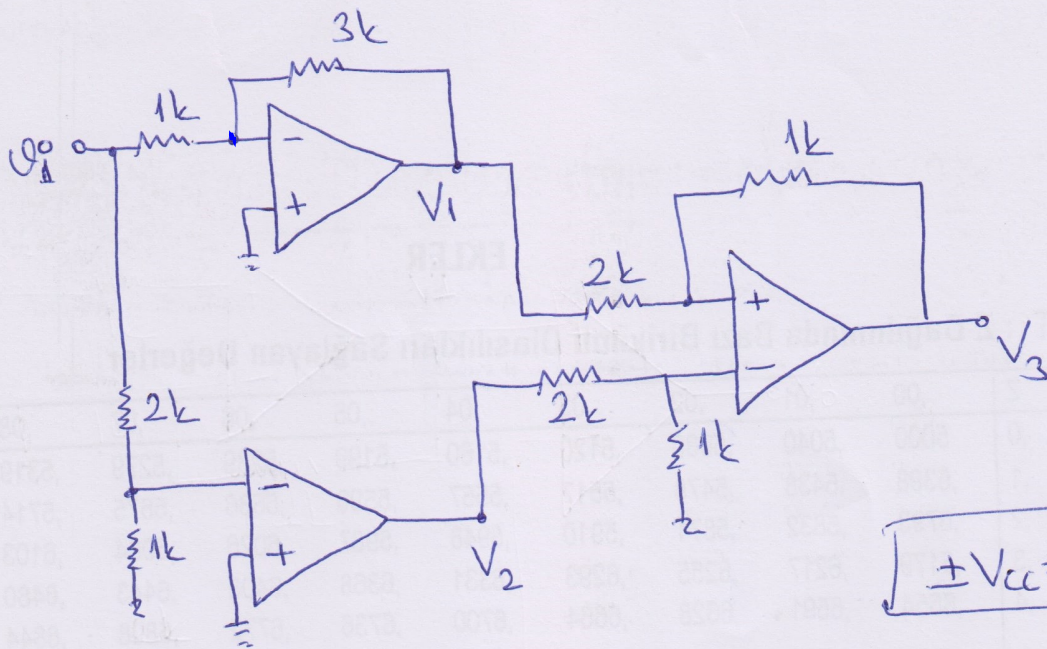
$$\frac{V_i - 0}{1k} = i$$

$$\frac{0 - V_2}{2k} = -i$$

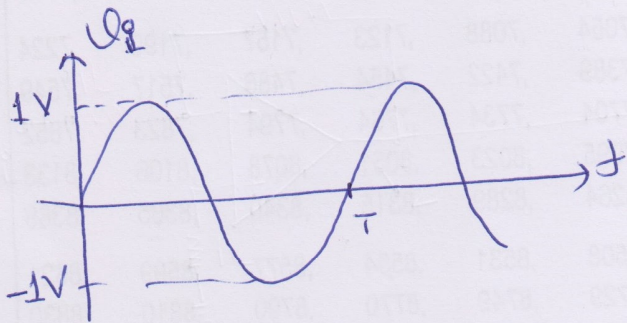
$$\frac{V_i}{1k} = i$$

şekildeki devrede V_i , $\pm 1V$ sinüzoidal bir sinyal ve tüm Opamp'lar için $\pm V_{cc} = \pm 6V$ olduğuna göre; V_1 , V_2 ve V_0 sinyallerini V_i 'ye göre ölçekli olarak çiziniz.

4



$$\pm V_{cc} = \pm 12V$$



Buna göre $V_1, V_2, V_3 = ?$ aızınız.